ModelIt!

Teacher's Guide

The Vaping Crisis: What's Really Happening to Your Lungs?

Modeling Chemical Damage to the Respiratory System

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model to illustrate how chemical exposure progressively damages the hierarchical organization of lung tissue.
Disciplinary Core Idea	**LS1.A Structure and Function** — Multicellular organisms have hierarchical structural organization, in which any one system is made up of numerous parts that interact to carry out life functions.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from chemical exposure to cellular damage to systemic inflammation and reduced lung function.

CAST Testing Alignment

MULTIPLE CHOICE:

A student has been vaping daily for one year. Which model prediction best explains why their lung function test shows a 15% decrease even though they feel fine?

CONSTRUCTED RESPONSE:

Using your model, explain the biological mechanism by which chronic vaping leads to progressive lung damage. Include the roles of inflammation and epithelial cell health in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The human respiratory system is a marvel of biological engineering — 300 million alveoli provide a gas exchange surface area the size of a tennis court. But this delicate tissue was never designed to handle the chemical cocktail delivered by e-cigarettes. Understanding the hierarchical organization of the lungs is essential for understanding how vaping causes damage at every level.

Respiratory System Organization

The respiratory system follows a hierarchical structure: the trachea branches into bronchi, which branch into bronchioles, which terminate in alveoli. Each level is lined with specialized epithelial cells. Alveoli are surrounded by capillaries where oxygen enters the blood and CO2 is removed. This organization maximizes surface area for gas exchange.

What's Actually in Vape Aerosol

E-cigarette aerosol is not water vapor. It contains nicotine, propylene glycol, vegetable glycerin, formaldehyde, acrolein, heavy metals (lead, nickel, chromium), and ultrafine particles. When heated, these chemicals undergo pyrolysis, creating new toxic compounds. The ultrafine particles penetrate deep into the alveoli where there is no mucociliary clearance.

Inflammation as a Double-Edged Sword

When chemicals damage lung cells, the immune system responds with inflammation — sending white blood cells, releasing cytokines, and increasing blood flow to the area. Short-term, this helps clear damage. But chronic inflammation causes collateral damage: healthy cells are destroyed, scar tissue forms, and the inflammatory molecules themselves damage surrounding tissue. This creates a self-reinforcing cycle of damage.

EVALI and Youth Vaping

In 2019, EVALI (E-cigarette or Vaping Product Use-Associated Lung Injury) hospitalized over 2,800 people and killed 68 in the US. Most were under 35. Even without acute EVALI, chronic vaping causes measurable reductions in lung function, increased respiratory symptoms, and cellular changes that may increase cancer risk. Adolescent lungs are especially vulnerable because they're still developing.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify nicotine concentration, chemical exposure duration, lung cell health, and inflammation response as the key components of the vaping damage system.

E **Establish Relationships**
Students determine that nicotine concentration decreases lung cell health, exposure duration increases inflammation, and inflammation further decreases cell health — creating a destructive feedback loop.

V **Visualize & Model**
Students build a computational model showing how chemical exposure leads to progressive lung damage through the inflammation-damage cycle.

E **Evaluate Outcomes**
Students run occasional use, daily heavy use, and cessation scenarios to compare short-term versus long-term outcomes and recovery potential.

R **Revise & Extend**
Students add oxygen exchange rate, mucociliary clearance, or immune cell activity to model more complete respiratory system dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In 2019, a healthy 17-year-old ended up on life support. The cause? Vaping. And he's not the only one."*

Watch For

Share a brief EVALI case study. Ask: How can something that seems harmless cause this much damage?

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling what actually happens inside your lungs when vape chemicals hit them."*

Watch For

Have students read objectives. Pre-teach 'inflammation response' and 'epithelial cells.'

Slide 3: Big Question

3 min

Say: *"Companies say it's harmless. Hospitals say otherwise. Let's find out who's right."*

Watch For

Anonymous poll: What do you think is in vape aerosol? Compare guesses to the actual chemical list.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of your actual lung cells being exposed to these chemicals."*

Watch For

Preview LEVER steps. Emphasize this is about biological mechanisms, not moral judgments.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key biological players when chemicals enter your lungs?"*

Watch For

Guide sorting. Discuss why nicotine concentration and duration are external (user-controlled).

Slide 6: Activity 2: Connections

8 min

Say: *"Here's where it gets scary — the body's own defense system starts causing MORE damage."*

Watch For

Help students identify the destructive feedback loop. Use red arrows for damage pathways.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's see what happens to lung cells after one month, six months, and two years."*

Watch For

Students predict damage curves first. Run simulations. Discuss the recovery scenario.

Slide 8: Discoveries

5 min

Say: *"Why didn't the occasional user's lungs recover completely? What does that tell us?"*

Watch For

Connect model findings to real clinical data. Discuss cumulative vs. reversible damage.

Slide 9: STEM Challenge

12 min

Say: *"The health department needs YOUR data to convince teens. Make it scientifically accurate AND compelling."*

Watch For

Groups design science-based campaigns using model evidence. Peer review for accuracy.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Nicotine Concentration, Chemical Exposure Duration
- **INTERNAL:** Lung Cell Health, Inflammation Response

Relationships

- **Nicotine Concentration to Lung Cell Health:** NEGATIVE (-) — Higher nicotine and chemical concentrations directly damage epithelial cells lining the airways. The chemicals cause oxidative stress, DNA damage, and cell death in the delicate lung tissue.
- **Chemical Exposure Duration to Inflammation Response:** POSITIVE (+) — Longer exposure duration means the immune system remains activated for extended periods. Chronic exposure leads to persistent, escalating inflammation that never fully resolves.
- **Inflammation Response to Lung Cell Health:** NEGATIVE (-) — Chronic inflammation damages healthy lung cells through collateral damage from immune molecules. The immune response meant to protect the lungs ends up accelerating tissue destruction.

Key Discoveries

- Nicotine and vaping chemicals directly damage epithelial cells, reducing their ability to protect the lungs and facilitate gas exchange
- Chronic inflammation creates a destructive cycle — the immune response meant to heal actually causes additional tissue damage over time
- The damage is cumulative and not always reversible — some lung cell damage becomes permanent after prolonged exposure
- Short-term effects are mild enough to ignore, which is why many vapers don't realize the damage until it's significant

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Vaping is just water vapor"	E-cigarette aerosol contains nicotine, heavy metals, volatile organic compounds, and ultrafine particles — not water vapor. The base liquids (propylene glycol and vegetable glycerin) break down into formaldehyde and acrolein when heated. These chemicals cause direct cellular damage.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If I don't feel sick, it's not hurting me"	Cumulative damage often has no symptoms until it's significant. Like high blood pressure or UV skin damage, vaping causes progressive cellular changes that aren't felt day-to-day but show up on medical tests and eventually as symptoms.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Vaping is safe because it's FDA-regulated"	E-cigarettes are regulated as tobacco products, not as safe products. The FDA has authorized very few e-cigarettes and only for adults already addicted to cigarettes as a harm-reduction tool — not as a safe alternative for non-smokers. No e-cigarettes have been approved as safe for youth.